

Java

Arduino IDE was officially built in Java environment and provides serial port connectivity through the RXTX Java library. With Java, you get access to numerous API and also attain access into their rich, dynamic interfaces. Java-Arduino library is used where Java API is then connected to Arduino through serial port either through a USB device or a wireless device. All the codes are managed internally for smooth communication between Java and Arduino.

C#

You need serial port communication to use C# in Arduino for connecting with your computer. CmdMessenger library goes further than checking your Arduino sketch and code in a different language altogether. Now this CmdMessenger can be run in Microsoft Visual Studio or any third party software to communicate using C# between your PC and Arduino. Programmers can call functions, send and receive commands to make their application work efficiently with coding. You can also use 3rd party devices like Netduino to run C# natively in its built *upon .NET* Micro Framework directly.

JavaScript

Maintained by the World Wide Consortium, JavaScript is one of the most used scripting languages, that combines with HTML and CSS for building interactive websites online. To accept JavaScript commands on Arduino, you need to install a Firmata protocol and connect it via a serial port. Firmata API runs on Arduino which the JavaScript calls to manipulate the code on the board itself.

Start by connecting Arduino to your laptop, PC or Mac with a USB cable or any wireless device and then fire up the Arduino IDE assuming you have it in the system already. Now go to Files > Examples > Firmata > StandardFirmata to upload the Standard Firmata to your Arduino files.

HTML/CSS/JavaScript

For working on any website development projects, these three languages work in combinations and are often called 'Front End Development.' These work on specific HTTP protocols to transfer the data from one server to the client-side.

Official Arduino IDE has many limitations, so it is recommended to use a third-party IDE such as Eclipse that can be easily plugged-in to support